

WORKING WITH GOOGLE EARTH ENGINE 2

TO SEARCH, FIND AND VISUALIZE A BROAD RANGE OF REMOTELY
SENSED DATASETS

Adding Multi-Image Data

SENTINEL 2 IMAGERIES

Key characteristics:

- 2 Satellites (S2A, S2B)
- Multi-spectral data with 13 bands
- Spatial resolution up to 10 m
- Revisit around five days from two-satellite A & B (at equator)
- Swath width = 290 km

Sentinel-2 Bands	Central Wavelength (μm)	Resolution (m)
Band 1 - Coastal aerosol	0.443	60
Band 2 - Blue	0.490	10
Band 3 - Green	0.560	10
Band 4 - Red	0.665	10
Band 5 - Vegetation Red Edge	0.705	20
Band 6 - Vegetation Red Edge	0.740	20
Band 7 - Vegetation Red Edge	0.783	20
Band 8 - NIR	0.842	10
Band 8A - Vegetation Red Edge	0.865	20
Band 9 - Water vapour	0.945	60
Band 10 - SWIR - Cirrus	1.375	60
Band 11 - SWIR	1.610	20
Band 12 - SWIR	2.190	20

Sentinel 2

Sentinel-2 MSI: MultiSpectral Instrument, Level-1C: [COPERNICUS/S2](#)

Earth Engine Snippet: `ee.ImageCollection("COPERNICUS/S2")`

The Sentinel-2 data contain 13 UINT16 (Unsigned Integer 16 bit) spectral bands representing TOA reflectance scaled by 10000.

Sentinel-2 MSI: MultiSpectral Instrument, Level-2A: [COPERNICUS/S2_SR](#)

Earth Engine Snippet: `ee.ImageCollection("COPERNICUS/S2_SR")`

The assets contain 12 UINT16 spectral bands representing SR scaled by 10000 (unlike in L1 data, there is no B10).

About Copernicus Sentinel-2...

WHAT?

A constellation of **two identical satellites in the same orbit**, Copernicus Sentinel-2 images land and coastal areas at high spatial resolution in the optical domain



WHICH?

Main applications include agriculture; land ecosystems monitoring; forests management; inland and coastal water quality monitoring; disasters mapping and civil security



WHEN?

Sentinel-2A was launched on 23 June 2015; Sentinel-2B on 7 March 2017, both on a Vega rocket from Kourou, French Guiana



DATA AND USERS

As of July 2020, about **20 million products** have been generated and made available for download, culminating a total of 10 Petabytes



DATA ACCESS

<https://scihub.copernicus.eu>

WHERE?

Designed and built by a group of around **60 companies** led by **Airbus Defence and Space** for the space segment and **Thales Alenia Space** for the ground segment



WHO?

Services include **CLMS** (Copernicus Land Monitoring Service); **CMEMS** (Copernicus Marine Environment Monitoring Service); **CEMS** (Copernicus Emergency Management Service) and Copernicus Security Service; among others



WHATS NEXT?

Continuity over the coming years will be ensured by the **launch of additional satellites** (Sentinel-2C and Sentinel-2D). Furthermore, a new generation of Sentinel-2 satellites is being prepared, to take up the relay from the first generation



Image Collection

- While working with DEM we dealt with mapping an 'Image'
- Now we will move to 'Image Collection' or collection of images as multi-band, multi temporal images.
- We will filter our required images based on their attributes- such as bands, dates, cloud cover, etc.

Importing Sentinel 2 Image Collection

1. Open a new code window to create a script
2. Find Sentinel-2 from the 'Data Catalogue'
3. Click on the surface reflectance datasets
4. Import using import button or copy the code and paste it in the code window.

```
var S2 = ee.ImageCollection("COPERNICUS/S2_SR");
```

5. `var S2` will contain the entire archive of images acquired from the beginning

Filter data for a specific period, or location, or quality

- From the entire set of the image collection we need to limit the number of images according to some criteria
 1. Filter by a period
 2. Filter by a location
 3. Filter by a metadata

FILTER IMAGE COLLECTION

**SELECTION FROM AN IMAGE COLLECTION BASED ON
CERTAIN CRITERIA**

Filter by a Period

`imageCollection.filterDate()`

EXERCISE: SELECT ONLY S2 IMAGES ACQUIRED IN 2020

1. Type the following code:

```
var S2_2020 = ee.ImageCollection("COPERNICUS/S2_SR")  
.filterDate('2020-01-01', '2020-08-07');
```

On the previous code, you may have noted that the method `filterDate` has been assigned to the selected image collection just concatenating the two functions with a dot. This is typical in GEE and allows to avoid to create multiple variables. Just be careful that the semicolon must be assigned just at the end of the functions.

Use the concatenation to create variables for each line of code. Below is how would appear the selection of images without concatenation:

```
var S2 = ee.ImageCollection("COPERNICUS/S2_SR");  
  
var S2_2020 = S2.filterDate('2020-01-01', '2020-08-07');
```

Filter by a Location

`imageCollection.filterBounds()`

FILTER BY A LOCATION

EXERCISE: RE-SELECT ONLY S2 IMAGES COVERING THE AOI

In most of the cases, you will also need to refine your selected images only to those that overlap specific geometries. This time, we will use a geometry digitized from the screen. Click on Draw a shape from the top-right tools of the map window:



Draw a polygon within the area to analyze. Click on the last vertex to close the geometry. Once the geometry is closed, you will see within the Import area, that the geometry appeared. You can now copy and paste the new geometry in your script. Then:

1. Rename the new geometry as “AOI”
2. Type the following code to refine the search:

```
filterBounds(AOI)
```

Note that `filterBounds` does not clip the images; it just selects those images that overlap the geometry.

Filter by a Metadata

```
imageCollection.filterMetadata()
```

EXERCISE: RE-SELECT ONLY S2 IMAGES WITH CLOUDS < 80%

1. Check the metadata of one image with the `.first()` method on the image collection
2. Type `filterMetadata` in the doc tab to understand which operator to use
3. Type the corresponding code to refine the selection

EXERCISE: COUNT THE NUMBER OF MATCHING IMAGES FROM EACH FILTER

In order to count how many individual images matches each filter, you can use the `size()` function after a filter is selected. You will see that after each filter is applied the resulting number of matching images will decrease.

Filter Image Collection

SELECT ONE IMAGE FROM AN IMAGE COLLECTION

Is mapping entire image collection possible?

Yes, theoretically possible but,

Adding an image collection to the map layer may end-up showing an error due to exceeding the computation time

FILTER THE SAME PERIOD OVER MULTIPLE YEARS

In some circumstances, it may be useful to select the same period across multiple years. While it is possible to create several image collections and then merge them, the `calendarRange()` function allows to accomplish this task easily. The function goes over the image collection and filter the images that are within the assigned range. The range can be expressed in various formats: e.g. year, month, day of year, ect.

To filter an image collection of images acquired in a specific month of a specific range of years, you can apply the following example:

```
var S2_TS = ee.ImageCollection("COPERNICUS/S2")
```

```
.filterDate('2010-01-01', '2020-01-01')
```

```
.filter(ee.Filter.calendarRange(7,7,'month'))
```

```
ee.Filter.calendarRange(start, end, field)
```

SELECT ONE IMAGE FROM AN IMAGE COLLECTION

Each image has its unique ID that can be used to access and display the specific image

Other options are:

- The `first()` function, returns the first image from the image collection.
- The `sort()` function returns the images in ascending or descending order, based on a property.
- Combining `sort` and `first`, you can return the most recent image matching your filters.

```
var mostRecent = S2_2020.sort('system:time_start', false).first();  
//False is in descending order
```


Example

Imports (1 entry) 

▸ var Studyarea: Polygon, 4 vertices  

```
var S2 = ee.ImageCollection("COPERNICUS/S2_SR")
```

```
//Six weeks after Karachi Lockdown --May 10, 2020, to June 6, 2020  
.filterDate('2020-05-10','2020-06-07')
```

```
//Select Malir Outlet at the Arabian Ocean  
.filterBounds(Studyarea)
```

```
//Filter for cloud cover  
.filterMetadata('CLOUDY_PIXEL_PERCENTAGE', 'less_than', 20)
```

```
//Select Blue, Green, Red and NIR bands with 10m resolution  
.select(['B2','B3','B4','B8'])
```

```
//clip image collection  
.map(function(image){return image.clip(Studyarea)});
```

```
Map.setCenter(67.10456998490511,24.78410264291859, 12);
```

```
print(S2);
```

Loading a Specific Sentinel-2 image

Assets format:

COPERNICUS/S2_SR/20151128T002653_20151128T102149_T56MNN.

Here

1. the first numeric part represents the sensing date and time
2. second numeric part represents the product generation date and time, and
3. the final 6-character string is a unique granule identifier indicating its UTM grid reference

Use this link

<https://code.earthengine.google.com/baeed711e6960beb8cf691c8e81bb6ec>

```
Image COPENICUS/S2/20180422T012719_20180422T012714_T52LHM (16 bands)
  type: Image
  id: COPENICUS/S2/20180422T012719_20180422T012714_T52LHM
  version: 1524595215620245
  ▶ bands: List (16 elements)
  ▶ properties: Object (62 properties)
```

<https://code.earthengine.google.com/2970513a29120fb7b6347ae3a9b2b07a>

<https://code.earthengine.google.com/4380b36d27ef37ed3b6060ad36567370>

```
image.normalizedDifference(['BANDx', 'BAND'])
```

Spectral Indices

```
var NDVI = image.normalizedDifference(['B8','B4']);
```

▼ ImageCollection COPENNICUS/S2_SR (17 elements)

JSON

type: ImageCollection

id: COPENNICUS/S2_SR

version: 1641260551661052

bands: []

▼ features: List (17 elements)

- ▶ 0: Image COPENNICUS/S2_SR/20200511T060629_20200511T062009_T42RTN (4 bands)
- ▶ 1: Image COPENNICUS/S2_SR/20200511T060629_20200511T062009_T42RUN (4 bands)
- ▶ 2: Image COPENNICUS/S2_SR/20200513T055641_20200513T060443_T42RTN (4 bands)
- ▶ 3: Image COPENNICUS/S2_SR/20200513T055641_20200513T060443_T42RUN (4 bands)
- ▶ 4: Image COPENNICUS/S2_SR/20200516T060641_20200516T061205_T42RTN (4 bands)
- ▶ 5: Image COPENNICUS/S2_SR/20200516T060641_20200516T061205_T42RUN (4 bands)
- ▶ 6: Image COPENNICUS/S2_SR/20200518T055639_20200518T060001_T42RTN (4 bands)
- ▶ 7: Image COPENNICUS/S2_SR/20200518T055639_20200518T060001_T42RUN (4 bands)
- ▶ 8: Image COPENNICUS/S2_SR/20200521T060629_20200521T061823_T42RTN (4 bands)
- ▶ 9: Image COPENNICUS/S2_SR/20200521T060629_20200521T061823_T42RUN (4 bands)
- ▶ 10: Image COPENNICUS/S2_SR/20200523T055641_20200523T060206_T42RTN (4 band...
- ▶ 11: Image COPENNICUS/S2_SR/20200523T055641_20200523T060206_T42RUN (4 band...
- ▶ 12: Image COPENNICUS/S2_SR/20200526T060641_20200526T061546_T42RTN (4 band...
- ▶ 13: Image COPENNICUS/S2_SR/20200526T060641_20200526T061546_T42RUN (4 band...
- ▶ 14: Image COPENNICUS/S2_SR/20200528T055639_20200528T060230_T42RTN (4 band...
- ▶ 15: Image COPENNICUS/S2_SR/20200528T055639_20200528T060230_T42RUN (4 band...
- ▶ 16: Image COPENNICUS/S2_SR/20200605T060641_20200605T061858_T42RTN (4 band...

▶ properties: Object (21 properties)

Applying reducers - applying stats

Mean = `ImageCollection.reduce(ee.Reducer.mean())`

OR `ImageCollection.mean()`

Median `.median()`

Minimum `.min()`

Maximum `.max()`

Lab work

Prepare a GEE code and follow these steps:

1. Select Sentinel 2 images. Use all THREE filters (select the metadata filter of your choice).
2. Select only one Sentinel 2 image from its image collection.
 - Map Individual Bands (B2, B3, B4, and B8)
 - Map True color combination
 - Map B8, B4, B3 combination
 - Map Index NDVI and use different color palette.

Coding Indices

// 1- Define variable NDSI from equation

```
var NDVI = image.expression(  
    "(NIR - RED) / (NIR + RED)",  
    {  
        RED: image.select("B4"), // RED  
        NIR: image.select("B8"), // NIR  
    }).rename('NDVI');  
Map.addLayer(NDVI, {min: -1, max: 1}, "NDVI");
```


Quiz

Add in the code (sentinel 2) you were writing the following spectral indices (search internet to get the formula) and add the indices layers in the map with appropriate visualization parameters (use import);

1. Normalized Difference Water Index (NWDI). + 1, -1
2. Normalized Difference Salinity Index (NDSI). +1, -1
3. Soil Adjusted Vegetation Index (SAVI). -1, +1
4. Advanced Vegetation Index (AVI). 0.00 to 114.98
5. Enhanced Vegetation Index (EVI). -1, +1
6. Difference Vegetation-Index (DVI). -1, +1

Import your shapefile and clip with all layers with it before adding the map layer.

Useful website

https://www.geospatialecology.com/intro_rs_lab2/

THANK YOU